Unit 1

Solutions

Nearly every process that goes on inside the body takes place in a liquid solution of one kind or another.

What this unit sets out to do

- explain how the different kinds of solutions come to be formed
- state the concentration of a solution in a range of units
- state Henry's law and Raoult's law, and explain what each one says
- tell ideal solutions apart from non-ideal ones
- account for the ways real solutions depart from Raoult's law
- describe colligative properties and link them to the molar mass of the solute
- explain why certain solutes show abnormal colligative properties in solution

Mixtures, not pure substances

- pure substances are rare in daily life; most materials mix two or more of them
- how useful a mixture is depends on its **composition**
- **brass** (Cu + Zn) behaves very differently from **German silver** (Cu + Zn + Ni) or **bronze** (Cu + Sn)
- fluoride ions in water: **1 ppm** guards against tooth decay; **1.5 ppm** mottles teeth; high levels are poisonous (sodium fluoride is a rat poison)
- intravenous injections are made up in water whose salt content matches the ionic concentration of blood plasma

Plan of the unit

- chiefly liquid solutions, and how they form
- then their properties: vapour pressure and colligative properties
- first, the kinds of solution; next, the different ways of stating how much solute a liquid solution holds

1.1 Types of Solutions

- a **solution** is a **homogeneous** mixture of two or more components: its make-up and properties are the same at every point
- the **solvent** is normally the component present in the greatest amount; it fixes the physical state of the solution
- every other component present is a **solute**
- this unit deals only with **binary solutions**, those made of two components

Binary solutions, by physical state

- two components; either one may be a solid, a liquid or a gas
- crossing solute state with solvent state gives nine kinds, set out in Table 1.A

Table 1.A Types of Solutions

Kind of solution	Solute	Solvent	Everyday example
Gaseous	Gas	Gas	oxygen and nitrogen gases mixed together
	Liquid	Gas	chloroform in nitrogen gas
	Solid	Gas	camphor in nitrogen gas
Liquid	Gas	Liquid	oxygen in water
	Liquid	Liquid	ethanol in water
	Solid	Liquid	glucose in water
Solid	Gas	Solid	hydrogen taken up by palladium
	Liquid	Solid	sodium amalgam (mercury with sodium)
	Solid	Solid	copper in gold

1.2 Expressing Concentration of Solutions

Describing composition

- a solution's make-up is given by stating its **concentration**
- this can be done **qualitatively** or **quantitatively**

Qualitative words

- **dilute**: comparatively very little solute
- **concentrated**: comparatively a great deal of solute
- such loose terms breed confusion in practice, so a numerical measure is needed

Concentration can be put in numbers in several ways; each is taken in turn below.

(i) Mass percentage (w/w)

$$\text{Mass \% of a component} = \frac{\text{mass of the component in the solution}}{\text{total mass of the solution}} \times 100$$

- 10% glucose in water by mass: 10 g glucose + 90 g water, giving 100 g of solution
- widely used in industrial chemical work
- commercial bleach: sodium hypochlorite in water at 3.62 mass percentage

(ii) Volume percentage (V/V)

$$\text{Volume \% of a component} = \frac{\text{volume of the component}}{\text{total volume of solution}} \times 100$$

- the share of the total solution volume taken up by one component

Volume percentage in use

- 10% ethanol in water: 10 mL ethanol made up with water to a total of 100 mL
- the usual unit for solutions of liquids
- car engine coolant: **ethylene glycol** antifreeze at 35% (v/v) brings the freezing point of water down to 255.4 K (–17.6 °C)

(iii) Mass by volume percentage (w/V)

- mass of solute dissolved in every 100 mL of solution
- a favourite unit in medicine and pharmacy

(iv) Parts per million (ppm)

$$\text{Parts per million} = \frac{\text{number of parts of the component}}{\text{total number of parts of all components of the solution}} \times 10^6$$

- handy when a solute is present only in **trace** amounts
- like percentages, may be mass/mass, volume/volume or mass/volume
- pollutant levels in water or air are often quoted in mg mL^{-1} or ppm
- 1 L of sea water weighs 1030 g and carries roughly 6×10^{-3} g of dissolved O_2
- that tiny amount is equally 5.8 g per 10^6 g of sea water, i.e. **5.8 ppm**

(v) Mole fraction (x)

- symbol x, with a right-hand subscript naming the component

$$\text{Mole fraction of a component} = \frac{\text{number of moles of the component}}{\text{total number of moles of all the components}}$$

- binary mixture holding n_A mol of A and n_B mol of B:

$$x_A = \frac{n_A}{n_A + n_B}$$

- solution of i components:

$$x_i = \frac{n_i}{n_1 + n_2 + \ldots + n_i} = \frac{n_i}{\Sigma\, n_i}$$

Mole fractions add to one

$$x_1 + x_2 + \ldots + x_i = 1$$

- true for any solution, however many components it has

Why mole fraction is useful

- ties physical properties such as vapour pressure to concentration
- convenient for calculations on mixtures of gases

Worked example mole fraction from mass %

A solution is 20% ethylene glycol ($C_2H_6O_2$) by mass. Find the mole fraction of ethylene glycol.

- take 100 g of solution (any starting amount gives the same answer): 20 g ethylene glycol and 80 g water
- molar mass of $C_2H_6O_2 = 12 \times 2 + 1 \times 6 + 16 \times 2 = 62\ \text{g mol}^{-1}$
- moles of $C_2H_6O_2 = \dfrac{20\ \text{g}}{62\ \text{g mol}^{-1}} = 0.322\ \text{mol}$ moles of water $= \dfrac{80\ \text{g}}{18\ \text{g mol}^{-1}} = 4.444\ \text{mol}$

$$x_{\text{glycol}} = \frac{\text{moles of } C_2H_6O_2}{\text{moles of } C_2H_6O_2 + \text{moles of } H_2O} = \frac{0.322\ \text{mol}}{0.322\ \text{mol} + 4.444\ \text{mol}} = \mathbf{0.068}$$

$$x_{\text{water}} = \frac{4.444\ \text{mol}}{0.322\ \text{mol} + 4.444\ \text{mol}} = \mathbf{0.932}$$

- check: since the two fractions sum to 1, $x_{\text{water}} = 1 - 0.068 = 0.932$

(vi) Molarity (M)

- moles of solute present in one litre (one cubic decimetre) **of solution**

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution in litre}}$$

- $0.25\ \text{mol L}^{-1}$ NaOH, also written 0.25 M: 0.25 mol of NaOH dissolved to make one litre (one cubic decimetre) of solution

Worked example molarity

5 g of NaOH is present in 450 mL of solution. What is its molarity?

- moles of NaOH $= \dfrac{5\ \text{g}}{40\ \text{g mol}^{-1}} = 0.125\ \text{mol}$
- volume in litres = 450 mL / $1000\ \text{mL L}^{-1}$
- substituting in the molarity relation:

$$\text{Molarity} = \frac{0.125\ \text{mol} \times 1000\ \text{mL L}^{-1}}{450\ \text{mL}} = \mathbf{0.278\ M} = 0.278\ \text{mol L}^{-1} = 0.278\ \text{mol dm}^{-3}$$

(vii) Molality (*m*)

- moles of solute per kilogram **of solvent**

$$\text{Molality } (m) = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$$

- 1.00 mol kg^{-1} (1.00 *m*) KCl: 1 mol (74.5 g) of KCl in 1 kg of water

Worked example molality

2.5 g of ethanoic acid (CH_3COOH) is dissolved in 75 g of benzene. Find the molality.

- molar mass of $C_2H_4O_2$ = 12 × 2 + 1 × 4 + 16 × 2 = 60 g mol^{-1}
- moles = $\frac{2.5\text{ g}}{60\text{ g mol}^{-1}}$ = 0.0417 mol
- benzene = 75 g / 1000 g kg^{-1} = 75 × 10^{-3} kg

$$m = \frac{0.0417\text{ mol} \times 1000\text{ g kg}^{-1}}{75\text{ g}} = \mathbf{0.556\text{ mol kg}^{-1}}$$

Choosing a unit: the effect of temperature

- every way of stating concentration has its own advantages and drawbacks
- **mass %, ppm, mole fraction and molality** do not change with temperature
- **molarity** does change with temperature, because a volume varies with temperature while a mass does not

1.3 Solubility

Definition

- **solubility**: the largest amount of a substance that will dissolve in a stated amount of solvent at a stated temperature

What it depends on

- the nature of both solute and solvent
- temperature and pressure; examined below for a solid, and for a gas, dissolving in a liquid

Intext questions

Q1 22 g of benzene (C_6H_6) is dissolved in 122 g of carbon tetrachloride (CCl_4). Work out the mass percentage of each.

Q2 A solution in carbon tetrachloride is 30% benzene by mass. Find the mole fraction of benzene.

Q3 Find the molarity of (a) 30 g of $Co(NO_3)_2.6H_2O$ in 4.3 L of solution; (b) 30 mL of 0.5 M H_2SO_4 after dilution to 500 mL.

Q4 What mass of urea (NH_2CONH_2) is needed to prepare 2.5 kg of a 0.25 molal aqueous solution?

Q5 Aqueous KI that is 20% (mass/mass) has a density of 1.202 g mL^{-1}. Find KI's (a) molality, (b) molarity and (c) mole fraction.

1.4 Solubility of a Solid in a Liquid

Like dissolves like

- not every solid dissolves in every liquid
- water readily dissolves sodium chloride and sugar, but not naphthalene or anthracene
- benzene does the reverse: it dissolves naphthalene and anthracene, not sodium chloride or sugar
- polar solutes go into polar solvents, non-polar solutes into non-polar solvents; a solute dissolves where the intermolecular interactions of solute and solvent resemble each other

Dissolution meets crystallisation

- **dissolution**: solid solute added to solvent passes into solution, raising its concentration
- **crystallisation**: dissolved particles strike the solid and leave the solution again
- eventually both run at the same rate: as many particles enter as leave, giving a **dynamic equilibrium**

$$\text{Solute} + \text{Solvent} \rightleftharpoons \text{Solution}$$

Saturated and unsaturated

- at equilibrium the solute concentration holds steady for the given temperature and pressure; gases dissolving in liquids behave the same way
- **saturated solution**: can take up no further solute at that temperature and pressure; it is in dynamic equilibrium with undissolved solute and holds the most solute a given amount of solvent can carry
- **unsaturated solution**: can still dissolve more solute at the same temperature
- the solute concentration in a saturated solution *is* the solubility

Effect of temperature

- besides the nature of the substances, temperature and pressure also govern solubility
- temperature changes alter the solubility of a solid in a liquid markedly
- the dissolution equilibrium is dynamic, so it obeys **Le Chatelier's principle**; for a nearly saturated solution:
- endothermic dissolving ($\Delta_{sol}H > 0$): solubility should **rise** as temperature rises
- exothermic dissolving ($\Delta_{sol}H < 0$): solubility should **fall**; experiment bears out both trends

Effect of pressure

- pressure makes no appreciable difference to how well a solid dissolves in a liquid
- the reason: solids and liquids hardly compress at all, so a change in pressure leaves them practically unaltered

1.5 Solubility of a Gas in a Liquid

Gases in water

- water dissolves many gases
- oxygen dissolves only slightly, yet that dissolved oxygen keeps all aquatic life going
- hydrogen chloride (HCl) gas, by contrast, is very soluble

Controlling factors

- how much gas a liquid dissolves depends strongly on **pressure**
- and also on temperature

Why pressure raises gas solubility

- a gas dissolves better in a liquid as the pressure on it goes up
- picture a solution with gas above it at pressure p and temperature T, in dynamic equilibrium: gas particles enter and leave the solution at the same rate
- squeeze the gas into a smaller volume: more gas particles per unit volume, so more of them strike the liquid surface and enter it
- more gas dissolves until a fresh equilibrium sets in; higher gas pressure above the solution, higher solubility

Figure 1.A Effect of pressure on the solubility of a gas. The concentration of dissolved gas is proportional to the pressure on the gas above the solution.

- **(a)** a vessel capped by a lightly loaded piston: gas sits over the solution, some gas particles are dissolved, the system is at equilibrium
- **(b)** extra load drives the piston down; the gas is compressed into less volume at higher pressure, and noticeably more gas particles are now dissolved in the liquid

Henry's law

- Henry first put pressure and gas solubility into a quantitative relation; Dalton, working in the same period, reached independently the conclusion that gas solubility in a liquid depends on the gas's partial pressure
- **statement:** at constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid or solution
- measuring solubility as mole fraction: the gas's mole fraction in solution is proportional to its partial pressure above the solution
- **usual form:** the partial pressure of the gas in the vapour phase (p) is proportional to its mole fraction in the solution (x)

$$p = K_H x \qquad K_H = \text{Henry's law constant}$$

Figure 1.B Experimental results for the solubility of HCl gas in cyclohexane at 293 K. The slope of the line is the Henry's Law constant, K_H.

- partial pressure of HCl (torr, up to about 1000) plotted against its mole fraction in cyclohexane (up to about 0.020)
- the measured points lie on a straight line through the origin, as $p = K_H x$ predicts

Reading K_H

- a graph of partial pressure against mole fraction of gas in solution gives a line like Figure 1.B
- at one temperature, different gases have different K_H (Table 1.B), so K_H depends on the nature of the gas
- from $p = K_H x$: at a given pressure, the **larger** the K_H, the **less** soluble the gas
- K_H for N_2 and O_2 grows as temperature rises, which shows that gases dissolve better as temperature falls

Cold water holds more gas

- gases grow more soluble as the temperature drops
- for this reason aquatic species are more at ease in cold water than in warm water

Henry's law at work

Soft drinks and soda water

- bottles are closed while under high pressure
- this pushes up the amount of CO_2 that dissolves

Scuba divers and the bends

- breathing air at high pressure underwater makes atmospheric gases more soluble in the blood, so divers carry high dissolved-gas levels
- on rising, pressure falls step by step and the dissolved gases come out, forming bubbles of nitrogen in the blood
- the bubbles block capillaries: the **bends**, painful and life-threatening

Worked example Henry's law

N_2 at a partial pressure of 0.987 bar is bubbled through water at 293 K. Given K_H for N_2 at 293 K = 76.48 kbar, how many millimoles of N_2 dissolve in 1 litre of water?

- solubility is tied to the gas's mole fraction in the solution, which Henry's law gives directly:

$$x(\text{nitrogen}) = \frac{p(\text{nitrogen})}{K_H} = \frac{0.987\ \text{bar}}{76{,}480\ \text{bar}} = 1.29 \times 10^{-5}$$

- 1 litre of water is 55.5 mol; with n mol of N_2 dissolved, and $n \ll 55.5$ so it is dropped from the denominator:

$$x(\text{nitrogen}) = \frac{n\ \text{mol}}{n\ \text{mol} + 55.5\ \text{mol}} \approx \frac{n}{55.5} = 1.29 \times 10^{-5}$$

$$n = 1.29 \times 10^{-5} \times 55.5\ \text{mol} = 7.16 \times 10^{-4}\ \text{mol} = \frac{7.16 \times 10^{-4}\ \text{mol} \times 1000\ \text{mmol}}{1\ \text{mol}} = \mathbf{0.716\ mmol}$$

Table 1.B Values of Henry's Law Constant for Some Selected Gases in Water

Gas	Temperature/K	K_H/kbar	Gas	Temperature/K	K_H/kbar
He	293	144.97	Argon	298	40.3
H_2	293	69.16	CO_2	298	1.67
N_2	293	76.48	Formaldehyde	298	1.83×10^{-5}
N_2	303	88.84	Methane	298	0.413
O_2	293	34.86	Vinyl chloride	298	0.611
O_2	303	46.82			

Diving tanks

- to prevent the bends and the toxic effect of too much nitrogen in the blood, scuba tanks carry air diluted with helium
- mix: 11.7% helium, 56.2% nitrogen, 32.1% oxygen

High altitude: anoxia

- oxygen's partial pressure is lower at altitude than at ground level
- people who live high up, and climbers, therefore have little oxygen in blood and tissues
- low blood oxygen leaves climbers weak and muddled in thought: the condition called **anoxia**

Effect of temperature on gas solubility

- gases become **less** soluble in liquids as temperature rises
- a dissolved gas is in the liquid phase, so dissolving resembles condensation and gives out heat
- dissolving is a dynamic equilibrium (seen for solids above) and must obey Le Chatelier's principle; being **exothermic**, it is pushed back by heating, so solubility drops

Intext questions

Q6 H_2S is a poisonous gas smelling of rotten eggs and is used in qualitative analysis. Its solubility in water at STP is 0.195 *m*. Work out its Henry's law constant.

Q7 At 298 K, K_H for CO_2 in water is 1.67×10^8 Pa. How much CO_2 is contained in 500 mL of soda water bottled under a CO_2 pressure of 2.5 atm at 298 K?

1.6 Vapour Pressure of Liquid Solutions

Liquid solutions

- formed whenever the solvent is a liquid; the solute may be a gas, liquid or solid
- gases in liquids were covered under solubility of a gas; here: liquids and solids in a liquid
- one or more components may be volatile; the solvent usually is, the solute may or may not be
- only binary solutions are treated: (i) liquid in liquid, (ii) solid in liquid

1.7 Vapour Pressure of Liquid–Liquid Solutions

- two volatile liquids, components 1 and 2, in a closed vessel: both evaporate until vapour and liquid reach equilibrium
- total vapour pressure p_{total}; partial vapour pressures p_1 and p_2
- p_1 and p_2 are linked to the mole fractions x_1 and x_2
- French chemist François Marie Raoult (1886) gave this link in quantitative form: **Raoult's law**

Raoult's law

- for a solution of volatile liquids, each component's partial vapour pressure is directly proportional to its mole fraction in the solution

component 1: $p_1 \propto x_1 \Rightarrow \boxed{p_1 = p_1^{\,0} x_1}$ component 2: $\boxed{p_2 = p_2^{\,0} x_2}$

- $p_1^{\,0}$, $p_2^{\,0}$ = vapour pressures of pure 1 and pure 2 at the same temperature

Total vapour pressure

- Dalton's law of partial pressures: the total pressure above the solution is the sum of the components' partial pressures

$$p_{\text{total}} = p_1 + p_2$$

- putting in Raoult's expressions:

$$p_{\text{total}} = x_1 p_1^{\,0} + x_2 p_2^{\,0} = (1 - x_2)\, p_1^{\,0} + x_2 p_2^{\,0}$$

$$\boxed{p_{\text{total}} = p_1^{\,0} + (p_2^{\,0} - p_1^{\,0})\, x_2}$$

What the last equation tells us

- (i) the total vapour pressure can be expressed through the mole fraction of just one component
- (ii) plotted against component 2's mole fraction, it gives a **straight line**
- (iii) as the mole fraction of component 1 grows, the total vapour pressure either falls or rises, depending on the pure components' vapour pressures

The straight-line plots

- p_1 or p_2 against x_1 and x_2 gives straight lines (I and II, Figure 1.C), each passing through the point where its mole fraction equals 1
- p_{total} against x_2 (line III) is also straight
- taking component 1 as the less volatile ($p_1^{\,0} < p_2^{\,0}$): p_{total} runs from a minimum of $p_1^{\,0}$ to a maximum of $p_2^{\,0}$
- the make-up of the vapour in equilibrium with the solution is fixed by the components' partial pressures

Figure 1.C The plot of vapour pressure and mole fraction of an ideal solution at constant temperature. The dashed lines I and II represent the partial pressure of the components. (It can be seen from the plot that p_1 and p_2 are directly proportional to x_1 and x_2, respectively). The total vapour pressure is given by line marked III in the figure.

- vapour pressure plotted across the full composition range, from pure 1 ($x_1 = 1$) to pure 2 ($x_2 = 1$)
- line I: p_1 falls from $p_1^{\,0}$ to zero; line II: p_2 climbs from zero to $p_2^{\,0}$
- line III, their sum, joins $p_1^{\,0}$ to $p_2^{\,0}$ in a straight line

Composition of the vapour

- let y_1, y_2 be the mole fractions of components 1 and 2 **in the vapour**
- Dalton's law of partial pressures then gives:

$$p_1 = y_1\,p_{\text{total}} \qquad p_2 = y_2\,p_{\text{total}} \qquad \text{in general} \quad \boxed{p_i = y_i\,p_{\text{total}}}$$

Worked example vapour pressure and vapour composition

At 298 K the vapour pressures of chloroform ($CHCl_3$) and dichloromethane (CH_2Cl_2) are 200 mm Hg and 415 mm Hg. A solution is made from 25.5 g of $CHCl_3$ and 40 g of CH_2Cl_2 at 298 K. Find (i) its vapour pressure and (ii) the mole fraction of each component in the vapour.

(i) Vapour pressure of the solution

- molar mass $CH_2Cl_2 = 12 \times 1 + 1 \times 2 + 35.5 \times 2 = 85$ g mol^{-1}
- molar mass $CHCl_3 = 12 \times 1 + 1 \times 1 + 35.5 \times 3 = 119.5$ g mol^{-1}
- moles $CH_2Cl_2 = \dfrac{40\text{ g}}{85\text{ g mol}^{-1}} = 0.47$ mol
- moles $CHCl_3 = \dfrac{25.5\text{ g}}{119.5\text{ g mol}^{-1}} = 0.213$ mol
- total $= 0.47 + 0.213 = 0.683$ mol
- $x(CH_2Cl_2) = 0.47\text{ mol} / 0.683\text{ mol} = 0.688$; $x(CHCl_3) = 1.00 - 0.688 = 0.312$
- from $p_{\text{total}} = p_1^{\,0} + (p_2^{\,0} - p_1^{\,0})\,x_2$:

$p_{\text{total}} = 200 + (415 - 200) \times 0.688 = 200 + 147.9$
$= \mathbf{347.9\text{ mm Hg}}$

(ii) Mole fractions in the vapour

- use $y_i = p_i/p_{\text{total}}$
- $p(CH_2Cl_2) = 0.688 \times 415\text{ mm Hg} = 285.5\text{ mm Hg}$
- $p(CHCl_3) = 0.312 \times 200\text{ mm Hg} = 62.4\text{ mm Hg}$
- $y(CH_2Cl_2) = 285.5\text{ mm Hg} / 347.9\text{ mm Hg} =$ **0.82**
- $y(CHCl_3) = 62.4\text{ mm Hg} / 347.9\text{ mm Hg} =$ **0.18**

The more volatile component enriches the vapour

- CH_2Cl_2 is the more volatile of the two ($p^0 = 415$ mm Hg against 200 mm Hg for $CHCl_3$)
- and the vapour is correspondingly richer in it ($y = 0.82$ against 0.18)
- conclusion: at equilibrium the vapour is always richer in the more volatile component

1.8 Raoult's Law as a Special Case of Henry's Law

Two laws side by side

Raoult: $p_i = x_i p_i^0$ Henry: $p = K_H x$

- for a gas dissolved in a liquid, one component is volatile enough to be a gas, and its solubility follows Henry's law
- in both laws, the partial pressure of the volatile component (or gas) is directly proportional to its mole fraction in solution

The only difference

- the proportionality constant: K_H in one, p_1^0 in the other
- so Raoult's law is Henry's law with K_H equal to p_1^0

1.9 Vapour Pressure of Solutions of Solids in Liquids

Solids dissolved in liquids

- examples: sodium chloride, glucose, urea or cane sugar in water; iodine or sulphur in carbon disulphide
- some of their physical properties differ markedly from those of the pure solvent, vapour pressure among them
- **vapour pressure**: a liquid evaporates at a given temperature, and at equilibrium the pressure its vapour exerts on the liquid is its vapour pressure

Why a solute lowers it

- in a pure liquid, solvent molecules occupy the whole surface
- with a **non-volatile** solute present, the vapour comes from the solvent only, and the solution's vapour pressure is found to be **lower** than the pure solvent's at the same temperature
- solute particles take up part of the surface, so less of it is solvent; fewer solvent molecules escape, and the vapour pressure falls

Figure 1.D Decrease in the vapour pressure of the solvent on account of the presence of solute in the solvent (a) evaporation of the molecules of the solvent from its surface is denoted by ●, (b) in a solution, solute particles have been denoted by ○ and they also occupy part of the surface area.

- **(a)** pure solvent in a closed container: the surface is made up entirely of solvent molecules, which escape into the vapour above
- **(b)** a solution of 1 mol of solute in 1 mol of solvent: solute particles sit among solvent molecules at the surface, so fewer solvent molecules are found in the vapour above

Amount matters, identity does not

- how far the solvent's vapour pressure drops depends on the **quantity** of non-volatile solute, not on what the solute is
- 1.0 mol of sucrose in 1 kg of water lowers the vapour pressure by nearly the same amount as 1.0 mol of urea in 1 kg of water at the same temperature

Raoult's law, general form

- whatever the solution, every volatile component has a partial vapour pressure in direct proportion to its mole fraction
- binary solution: solvent = 1, solute = 2; with a non-volatile solute, only solvent molecules are in the vapour and supply the vapour pressure, p_1

Raoult's law for the solvent

- solvent's vapour pressure p_1, mole fraction x_1, vapour pressure when pure p_1^0

$$p_1 \propto x_1 \Rightarrow \boxed{p_1 = x_1 p_1^0}$$

- the proportionality constant is the pure solvent's vapour pressure, p_1^0
- vapour pressure against solvent mole fraction is a straight line (Figure 1.E)

Figure 1.E If a solution obeys Raoult's law for all concentrations, its vapour pressure would vary linearly from zero to the vapour pressure of the pure solvent.

- vapour pressure against mole fraction of solvent: a straight line rising from zero to p_1^0

1.10 Ideal and Non-ideal Solutions

Raoult's law is the yardstick for sorting liquid–liquid solutions into two classes: ideal and non-ideal.

1.11 Ideal Solutions

Definition and properties

- **ideal solution**: obeys Raoult's law at every concentration
- mixing the pure components involves no enthalpy change and no volume change:

$$\Delta_{mix}H = 0, \quad \Delta_{mix}V = 0$$

- no heat is taken in or given out on mixing, and the solution's volume equals the sum of the two volumes

Molecular picture

- pure A and pure B have A–A and B–B attractions; the solution has A–B attractions as well
- when A–A and B–B forces are close to A–B forces, the solution is ideal
- perfect ideality is rare; nearly ideal: n-hexane + n-heptane, bromoethane + chloroethane, benzene + toluene

1.12 Non-ideal Solutions

Definition

- **non-ideal solution**: fails to obey Raoult's law across the whole concentration range
- its vapour pressure is above or below the Raoult's law value $p_{total} = p_1^0 + (p_2^0 - p_1^0)x_2$
- higher: **positive deviation**; lower: **negative deviation**; plots in Figure 1.F

Positive deviation: the cause

- the deviations stem from the kind of interactions between molecules
- A–B attractions (solute–solvent) are **weaker** than A–A and B–B (solute–solute, solvent–solvent)
- molecules of A or B then escape more easily than from the pure liquid, which raises the vapour pressure

Figure 1.F The vapour pressures of two component systems as a function of composition (a) a solution that shows positive deviation from Raoult's law and (b) a solution that shows negative deviation from Raoult's law.

- each panel plots vapour pressure against mole fraction, from $x_1 = 0$ ($x_2 = 1$) to $x_1 = 1$ ($x_2 = 0$); dashed straight lines mark what Raoult's law predicts for p_1, p_2 and the total
- **(a)** the real curves for p_1, p_2 and the solution's vapour pressure bow **above** the dashed lines
- **(b)** the real curves sag **below** the dashed lines

Positive deviation: examples

- **ethanol + acetone**: pure ethanol is hydrogen bonded; acetone molecules push in between ethanol molecules and break some of these bonds; the weakened interactions give positive deviation (Figure 1.F a)
- **acetone + carbon disulphide**: solute–solvent dipolar interactions are weaker than the solute–solute and solvent–solvent ones, so this too deviates positively

Negative deviation: the cause

- A–A and B–B attractions are **weaker** than A–B
- vapour pressure drops, giving negative deviation
- **phenol + aniline**: the hydrogen bond between phenol's proton and the lone pair on aniline's nitrogen is stronger than the hydrogen bonds each liquid forms with itself

Chloroform + acetone

- also deviates negatively from Raoult's law
- a chloroform molecule can hydrogen-bond to an acetone molecule (Figure 1.G)
- each component's escaping tendency is reduced, vapour pressure falls: negative deviation (Figure 1.F b)

Figure 1.G

$$(H_3C)_2C{=}O \cdots H{-}CCl_3$$

- the H atom of chloroform ($CHCl_3$) is hydrogen bonded to the O atom of acetone's C=O group

Azeotropes

- **azeotrope**: a binary mixture whose liquid and vapour have the same composition, and which boils at a constant temperature
- its components therefore cannot be separated by fractional distillation
- two kinds: **minimum boiling azeotrope** and **maximum boiling azeotrope**
- a strongly **positive** departure from Raoult's law produces, at one particular composition, a minimum boiling azeotrope

Minimum boiling: ethanol–water

- ethanol–water from fermenting sugars, on fractional distillation, yields about 95% ethanol by volume
- at this **azeotrope composition** liquid and vapour match, and no further separation takes place

Maximum boiling: nitric acid–water

- formed at a specific composition by solutions with a large **negative** deviation from Raoult's law
- nitric acid–water: about 68% nitric acid and 32% water by mass, boiling at 393.5 K

Intext question

Q8 At 350 K, pure liquids A and B have vapour pressures of 450 and 700 mm Hg. A mixture of them has a total vapour pressure of 600 mm Hg. What is the composition of the liquid, and of the vapour in equilibrium with it?

1.13 Colligative Properties and Determination of Molar Mass

Four linked properties

- a non-volatile solute lowers a volatile solvent's vapour pressure; several properties are tied to this drop:
- (1) relative lowering of the solvent's vapour pressure; (2) depression of the solvent's freezing point; (3) elevation of the solvent's boiling point; (4) osmotic pressure of the solution

Colligative properties

- all four depend on the number of solute particles relative to the total particles in solution, whatever those particles are
- such properties are **colligative** (Latin *co*, together; *ligare*, to bind); each is taken up in turn

1.14 Relative Lowering of Vapour Pressure

- a solvent's vapour pressure in solution is below that of the pure solvent; Raoult found the lowering depends only on the concentration of solute particles, not on their identity
- Raoult's law relates the solution's vapour pressure, the mole fraction and the solvent's vapour pressure:

$$p_1 = x_1\, p_1^{\,0}$$

- the fall in the solvent's vapour pressure, Δp_1, is then:

$$\Delta p_1 = p_1^{\,0} - p_1 = p_1^{\,0} - p_1^{\,0} x_1 = p_1^{\,0}(1 - x_1)$$

- since $x_2 = 1 - x_1$:

$$\boxed{\Delta p_1 = x_2\, p_1^{\,0}}$$

- with several non-volatile solutes, the lowering depends on the **sum** of their mole fractions; rearranged:

$$\frac{\Delta p_1}{p_1^{\,0}} = \frac{p_1^{\,0} - p_1}{p_1^{\,0}} = x_2$$

Relative lowering of vapour pressure

- the left-hand side, $\Delta p_1/p_1^{\,0}$, is the **relative lowering of vapour pressure**; it equals the solute's mole fraction
- n_1, n_2 = moles of solvent and solute in the solution:

$$\frac{p_1^{\,0} - p_1}{p_1^{\,0}} = \frac{n_2}{n_1 + n_2} \quad (\text{as } x_2 = n_2/(n_1 + n_2))$$

- dilute solution: $n_2 \ll n_1$, so n_2 is dropped from the denominator:

$$\frac{p_1^{\,0} - p_1}{p_1^{\,0}} = \frac{n_2}{n_1}$$

- in terms of masses w_1, w_2 and molar masses M_1, M_2 of solvent and solute:

$$\frac{p_1^{\,0} - p_1}{p_1^{\,0}} = \frac{w_2 \times M_1}{M_2 \times w_1}$$

- with every other quantity known, this yields the solute's molar mass M_2

Worked example molar mass from vapour-pressure lowering

Pure benzene (molar mass 78 g mol^{-1}) has a vapour pressure of 0.850 bar at a certain temperature. Adding 0.5 g of a non-volatile non-electrolyte solid to 39.0 g of benzene brings the vapour pressure to 0.845 bar. Find the solid's molar mass.

- data: $p_1^{\,0} = 0.850$ bar; $p = 0.845$ bar; $M_1 = 78$ g mol^{-1}; $w_2 = 0.5$ g; $w_1 = 39$ g
- substituting in the mass form of the relative lowering relation:

$$\frac{0.850\ \text{bar} - 0.845\ \text{bar}}{0.850\ \text{bar}} = \frac{0.5\ \text{g} \times 78\ \text{g mol}^{-1}}{M_2 \times 39\ \text{g}} \Rightarrow \mathbf{M_2 = 170\ g\ mol^{-1}}$$

1.15 Elevation of Boiling Point

When a liquid boils

- vapour pressure rises as temperature rises
- a liquid boils at the temperature where its vapour pressure matches atmospheric pressure
- water boils at 373.15 K (100 °C) because its vapour pressure there is 1.013 bar (1 atmosphere)

Why a solution boils higher

- a non-volatile solute lowers the solvent's vapour pressure; Figure 1.H follows vapour pressure against temperature for pure solvent and solution
- aqueous sucrose at 373.15 K has a vapour pressure below 1.013 bar
- to boil, it must be heated above the boiling point of pure water until its vapour pressure reaches 1.013 bar

Boiling point elevation

- a solution always boils at a higher temperature than the pure solvent it is made from (Figure 1.H)
- as with vapour-pressure lowering, the rise depends on the number of solute molecules, not their kind
- 1 mol of sucrose in 1000 g of water boils at 373.52 K under one atmosphere
- T_b^0 = boiling point of pure solvent, T_b = of solution; **elevation of boiling point** $\Delta T_b = T_b - T_b^0$

Figure 1.H The vapour pressure curve for solution lies below the curve for pure water. The diagram shows that ΔT_b denotes the elevation of boiling point of a solvent in solution.

- vapour pressure against temperature: two rising curves, the solution's lying under pure water's
- a horizontal line at 1.013 bar (1 atm) cuts pure water's curve at T_b^0 and the solution's at the higher T_b; the gap between them is ΔT_b

ΔT_b and molality

- experiment shows that for dilute solutions ΔT_b is directly proportional to the solute's molal concentration:

$$\Delta T_b \propto m \Rightarrow \boxed{\Delta T_b = K_b\, m}$$

- m (molality) = moles of solute per 1 kg of solvent
- K_b = **Boiling Point Elevation Constant**, also called **Molal Elevation Constant** or **Ebullioscopic Constant**; unit K kg mol^{-1}; values for common solvents in Table 1.C
- w_2 g of solute (molar mass M_2) in w_1 g of solvent:

$$m = \frac{w_2/M_2}{w_1/1000} = \frac{1000 \times w_2}{M_2 \times w_1}$$

$$\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1}$$

- rearranged for the molar mass:

$$\boxed{M_2 = \frac{K_b \times 1000 \times w_2}{\Delta T_b \times w_1}}$$

- method: dissolve a known mass of solute in a known mass of a solvent of known K_b, and measure ΔT_b

Worked example boiling point of a solution

18 g of glucose, $C_6H_{12}O_6$, is dissolved in 1 kg of water in a saucepan. At what temperature does this water boil at 1.013 bar? (K_b for water = 0.52 K kg mol^{-1})

- moles of glucose = 18 g / 180 g mol^{-1} = 0.1 mol; solvent = 1 kg, so molality = 0.1 mol kg^{-1}
- $\Delta T_b = K_b \times m$ = 0.52 K kg mol^{-1} × 0.1 mol kg^{-1} = 0.052 K
- pure water boils at 373.15 K at 1.013 bar, so the solution boils at 373.15 + 0.052 = **373.202 K**

Worked example molar mass from ΔT_b

Benzene boils at 353.23 K. Dissolving 1.80 g of a non-volatile solute in 90 g of benzene lifts the boiling point to 354.11 K. With K_b for benzene = 2.53 K kg mol^{-1}, find the solute's molar mass.

- ΔT_b = 354.11 K − 353.23 K = 0.88 K
- substituting in $M_2 = K_b \times 1000 \times w_2 / (\Delta T_b \times w_1)$:

$$M_2 = \frac{2.53\ \text{K kg mol}^{-1} \times 1.8\ \text{g} \times 1000\ \text{g kg}^{-1}}{0.88\ \text{K} \times 90\ \text{g}} = \mathbf{58\ g\ mol^{-1}}$$

1.16 Depression of Freezing Point

What the freezing point is

- lower vapour pressure in a solution means a lower freezing point than the pure solvent's (Figure 1.I)
- at the freezing point, solid and liquid are in dynamic equilibrium
- so the **freezing point** is the temperature at which the liquid's vapour pressure equals the solid's vapour pressure

Why a solution freezes lower

- a solution freezes when its vapour pressure equals that of the pure solid solvent
- by Raoult's law, a non-volatile solid solute lowers the solvent's vapour pressure
- that lower vapour pressure matches the solid solvent's only at a lower temperature, so the freezing point drops

Figure 1.I Diagram showing ΔT_f, depression of the freezing point of a solvent in a solution.

- vapour pressure against temperature for the solid solvent, the pure liquid solvent and the solution; the solution's curve lies below the pure liquid's
- the pure liquid's curve meets the solid's curve at T_f^0; the solution's curve meets it at the lower T_f; the gap is ΔT_f

ΔT_f and molality

- T_f^0 = freezing point of pure solvent; T_f = freezing point with non-volatile solute dissolved in it
- **depression in freezing point**: $\Delta T_f = T_f^0 - T_f$
- as for boiling point elevation, in a dilute (ideal) solution ΔT_f is directly proportional to the molality m:

$$\Delta T_f \propto m \Rightarrow \boxed{\Delta T_f = K_f\, m}$$

- K_f depends on the nature of the solvent; it is the **Freezing Point Depression Constant**, or Molal Depression Constant

K_f and the molar mass relation

- K_f is also known as the **Cryoscopic Constant**; unit K kg mol^{-1}; values for common solvents in Table 1.C
- w_2 g of solute of molar mass M_2, in w_1 g of solvent, depresses the freezing point by ΔT_f; its molality is, as before, $m = (w_2/M_2)/(w_1/1000)$
- put into $\Delta T_f = K_f m$:

$$\Delta T_f = \frac{K_f \times w_2/M_2}{w_1/1000} = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1}$$

- solved for the molar mass:

$$M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$

- so the molar mass needs w_1, w_2, ΔT_f and the molal freezing point depression constant

Where K_f and K_b come from

- both depend on the solvent and can be worked out from:

$$K_f = \frac{R \times M_1 \times T_f^2}{1000 \times \Delta_{fus}H} \qquad K_b = \frac{R \times M_1 \times T_b^2}{1000 \times \Delta_{vap}H}$$

- R = gas constant; M_1 = molar mass of solvent
- T_f, T_b = freezing and boiling points of the pure solvent, in kelvin
- $\Delta_{fus}H$, $\Delta_{vap}H$ = enthalpies of fusion and of vapourisation of the solvent

TABLE 1.C Molal Boiling Point Elevation and Freezing Point Depression Constants for Some Solvents

Solvent	b. p./K	K_b/K kg mol^{-1}	f. p./K	K_f/K kg mol^{-1}
Water	373.15	0.52	273.0	1.86
Ethanol	351.5	1.20	155.7	1.99
Cyclohexane	353.74	2.79	279.55	20.00
Benzene	353.3	2.53	278.6	5.12
Chloroform	334.4	3.63	209.6	4.79
Carbon tetrachloride	350.0	5.03	250.5	31.8
Carbon disulphide	319.4	2.34	164.2	3.83
Diethyl ether	307.8	2.02	156.9	1.79
Acetic acid	391.1	2.93	290.0	3.90

Worked example freezing point of an antifreeze solution

45 g of ethylene glycol ($C_2H_6O_2$) is mixed with 600 g of water. Find (a) the depression of freezing point and (b) the freezing point of the solution.

- freezing point depression depends on molality, so find the molality of ethylene glycol = moles of ethylene glycol / mass of water in kg
- moles of ethylene glycol = $\frac{45\ \text{g}}{62\ \text{g mol}^{-1}} = 0.73\ \text{mol}$; mass of water = $\frac{600\ \text{g}}{1000\ \text{g kg}^{-1}} = 0.6\ \text{kg}$
- molality = $\frac{0.73\ \text{mol}}{0.60\ \text{kg}} = 1.2\ \text{mol kg}^{-1}$
- (a) $\Delta T_f = 1.86\ \text{K kg mol}^{-1} \times 1.2\ \text{mol kg}^{-1} =$ **2.2 K**
- (b) freezing point = 273.15 K – 2.2 K = **270.95 K**

Worked example molar mass from ΔT_f

Dissolving 1.00 g of a non-electrolyte in 50 g of benzene lowers benzene's freezing point by 0.40 K. Benzene's K_f is 5.12 K kg mol^{-1}. What is the solute's molar mass?

- substituting in $M_2 = K_f \times w_2 \times 1000 / (\Delta T_f \times w_1)$:

$$M_2 = \frac{5.12\ \text{K kg mol}^{-1} \times 1.00\ \text{g} \times 1000\ \text{g kg}^{-1}}{0.40 \times 50\ \text{g}} = \mathbf{256\ g\ mol^{-1}}$$

1.17 Osmosis and Osmotic Pressure

Everyday observations

- unripe mangoes kept in brine (salt water) to pickle them wrinkle and shrink
- wilted flowers perk up when stood in fresh water
- placed in salty water, blood cells shrink and collapse
- the common thread: in every case the substance is enclosed by a **membrane**

Membranes

- may be of animal or vegetable origin and occur naturally (pig's bladder, parchment), or be synthetic (cellophane)
- they look like unbroken sheets or films, yet are threaded with a network of submicroscopic holes or pores

Semipermeable membranes and osmosis

- small solvent molecules such as water pass through the pores; larger solute molecules are held back
- membranes that behave like this are **semipermeable membranes (SPM)**
- with such a membrane (passing solvent only) between solvent and solution (Figure 1.J), solvent flows through it from the pure solvent into the solution
- this flow of solvent is **osmosis**; it goes on until equilibrium is reached

Osmotic pressure

- applying some extra pressure to the solution can halt the flow of solvent into it
- the pressure that **just** stops the flow is the **osmotic pressure** of the solution
- osmosis drives solvent from dilute to concentrated solution across the membrane: always from lower to higher concentration
- osmotic pressure depends on the solution's concentration

Figure 1.J Level of solution rises in the thistle funnel due to osmosis of solvent.

- a solution in a thistle funnel whose mouth is closed by a semipermeable membrane, dipped into pure solvent
- solvent passes in through the membrane, and the solution column climbs up the funnel stem

Figure 1.K The excess pressure equal to the osmotic pressure must be applied on the solution side to prevent osmosis.

- solvent and solution separated by a semipermeable membrane; a pressure pushes on the solution side
- when that pressure equals the osmotic pressure, no solvent crosses the membrane

Definition and formula

- **osmotic pressure**: the excess pressure that has to be applied to a solution to prevent osmosis, that is, to stop solvent molecules moving through a semipermeable membrane into it (Figure 1.K)
- colligative: it depends on the number of solute molecules, not on their identity
- experiment shows that for dilute solutions Π is proportional to the molarity C at a given temperature T:

$$\boxed{\Pi = C\,R\,T} \quad (\Pi = \text{osmotic pressure}, R = \text{gas constant}) \qquad \Pi = (n_2/V)\,R\,T$$

- V = volume of solution in litres holding n_2 moles of solute; for w_2 g of solute of molar mass M_2, $n_2 = w_2/M_2$:

$$\Pi V = \frac{w_2\,R\,T}{M_2} \quad \text{or} \quad \boxed{M_2 = \frac{w_2\,R\,T}{\Pi V}}$$

- knowing w_2, T, Π and V gives the molar mass of the solute
- osmotic pressure measurement is thus a further route to molar masses, widely used for proteins, polymers and other macromolecules

Why the osmotic method is favoured

- pressure is measured at about room temperature
- it uses molarity rather than molality
- the effect is large even for very dilute solutions, unlike the other colligative properties
- especially suited to biomolecules, which are generally unstable at higher temperatures, and to polymers, which dissolve poorly

Isotonic, hypertonic, hypotonic

- **isotonic**: two solutions with equal osmotic pressure at a given temperature; separated by a semipermeable membrane, they show no osmosis
- fluid inside blood cells matches 0.9% (mass/volume) sodium chloride, **normal saline**, which is safe to inject intravenously
- **hypertonic**: above 0.9% (mass/volume) NaCl; water leaves the cells and they shrink
- **hypotonic**: below 0.9% (mass/volume) NaCl; water enters the cells and they swell

Worked example molar mass of a protein

An aqueous protein solution of 200 cm^3 holds 1.26 g of the protein and has an osmotic pressure of 2.57×10^{-3} bar at 300 K. Find the protein's molar mass.

- data: $\Pi = 2.57 \times 10^{-3}$ bar; $V = 200\ cm^3 = 0.200$ litre; $T = 300$ K; $R = 0.083$ L bar mol^{-1} K^{-1}
- substituting in $M_2 = w_2RT/\Pi V$:

$$M_2 = \frac{1.26\ \text{g} \times 0.083\ \text{L bar K}^{-1}\ \text{mol}^{-1} \times 300\ \text{K}}{2.57 \times 10^{-3}\ \text{bar} \times 0.200\ \text{L}} = \mathbf{61{,}022\ g\ mol^{-1}}$$

Osmosis explains the everyday observations

- a raw mango in concentrated brine loses water by osmosis and shrivels into pickle
- wilted flowers recover in fresh water
- a carrot gone limp from losing water to the air firms up again in water, as water moves into its cells by osmosis
- blood cells in water with less than 0.9% (mass/volume) salt swell as water flows in by osmosis
- eating a lot of salt or salty food makes water collect, by osmosis, in tissue cells and in the spaces between cells

More osmosis in daily life

- the puffiness or swelling from water held in tissues is **edema**
- water rising from soil into roots and on up the plant is partly due to osmosis
- salting meat and sugaring fruit guard against bacteria: on salted meat or candied fruit a bacterium loses water by osmosis, shrivels and dies

Figure 1.L Reverse osmosis occurs when a pressure larger than the osmotic pressure is applied to the solution.

- a salt-water chamber and a fresh-water chamber divided by a semipermeable membrane (SPM); a piston presses on the salt water with a pressure greater than Π
- pure water is driven across the membrane into the fresh-water chamber and leaves by an outlet

1.18 Reverse Osmosis and Water Purification

Reversing the flow

- press on the solution side with more than the osmotic pressure, and osmosis runs backwards: pure solvent leaves the solution through the semipermeable membrane
- this is **reverse osmosis**, a process of great practical value
- used to **desalinate sea water** (Figure 1.L): above osmotic pressure, pure water is squeezed out of sea water through the membrane

Membranes for the job

- many polymer membranes are available; the pressure needed is quite high
- a practical porous membrane: a cellulose acetate film on a suitable support
- cellulose acetate lets water through but not the impurities and ions of sea water
- many countries now rely on desalination plants for drinking water

Intext questions

Q9 Pure water at 298 K has a vapour pressure of 23.8 mm Hg. If 50 g of urea (NH_2CONH_2) is dissolved in 850 g of water, what are the vapour pressure of water over this solution and its relative lowering?

Q10 At 750 mm Hg water boils at 99.63 °C. What amount of sucrose must be added to 500 g of water to make it boil at 100 °C?

Q11 What mass of ascorbic acid (Vitamin C, $C_6H_8O_6$) must be dissolved in 75 g of acetic acid to lower its melting point by 1.5 °C? K_f = 3.9 K kg mol^{-1}.

Q12 1.0 g of a polymer of molar mass 185,000 is dissolved in 450 mL of water at 37 °C. Find the osmotic pressure, in pascals.

1.19 Abnormal Molar Masses

Dissociation of ionic solutes

- ionic compounds split into cations and anions in water: 1 mol of KCl (74.5 g) should release 1 mol of K^+ and 1 mol of Cl^-, i.e. 2 mol of particles
- ignoring interionic attractions, 1 mol of KCl in 1 kg of water should raise the boiling point by 2 × 0.52 K = 1.04 K

Dissociation lowers the apparent molar mass

- unaware of the dissociation, one would take 74.5 g as the mass of 2 mol of particles, and so put the molar mass of KCl at 37.25 g
- general rule: when a solute dissociates into ions, the experimental molar mass always comes out **lower** than the true value

Association raises it

- ethanoic (acetic) acid molecules pair up in benzene through hydrogen bonding, as is usual in solvents of low dielectric constant; dimerisation cuts the number of particles

$$2\ CH_3COOH \rightleftharpoons (CH_3COOH)_2$$

- if every molecule associates, ΔT_b or ΔT_f is half the normal value, and the molar mass calculated from it is **twice** the expected one

A molar mass that turns out lower or higher than the expected (normal) value is called an **abnormal molar mass**.

The van't Hoff factor, *i*

- brought in by van't Hoff in 1880 to allow for the extent of association or dissociation:

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}} = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$$

$$i = \frac{\text{Total number of moles of particles after association/dissociation}}{\text{Number of moles of particles before association/dissociation}}$$

- abnormal molar mass = the molar mass found by experiment
- calculated colligative property = the value assuming the non-volatile solute neither associates nor dissociates
- association: $i < 1$; dissociation: $i > 1$
- aqueous KCl: i close to 2; ethanoic acid in benzene: i nearly 0.5

Colligative equations with *i* included

- relative lowering of vapour pressure of solvent: $\frac{p_1^{\,0} - p_1}{p_1^{\,0}} = i \cdot \frac{n_2}{n_1}$
- elevation of boiling point: $\Delta T_b = i\,K_b\,m$
- depression of freezing point: $\Delta T_f = i\,K_f\,m$
- osmotic pressure of solution: $\Pi = i\,n_2\,R\,T\,/\,V$

Worked example degree of association (part 1)

2 g of benzoic acid (C_6H_5COOH) in 25 g of benzene depresses the freezing point by 1.62 K. Benzene's molal depression constant is 4.9 K kg mol^{-1}. If the acid dimerises in solution, what percentage of it is associated?

- data: w_2 = 2 g; K_f = 4.9 K kg mol^{-1}; w_1 = 25 g; ΔT_f = 1.62 K
- from $M_2 = K_f \times w_2 \times 1000 / (\Delta T_f \times w_1)$:

$$M_2 = \frac{4.9\ \text{K kg mol}^{-1} \times 2\ \text{g} \times 1000\ \text{g kg}^{-1}}{25\ \text{g} \times 1.62\ \text{K}} = 241.98\ \text{g mol}^{-1}$$ (experimental molar mass in benzene)

$$2\ C_6H_5COOH \rightleftharpoons (C_6H_5COOH)_2$$

- with degree of association x: $(1 - x)$ mol stays unassociated and $x/2$ mol of dimer forms at equilibrium
- total moles of particles at equilibrium = $1 - x + x/2 = 1 - x/2$, and this equals the van't Hoff factor i
- but also i = normal molar mass / abnormal molar mass

Measured i for strong electrolytes

- values of i for a few strong electrolytes appear in Table 1.D
- for KCl, NaCl and $MgSO_4$, i tends towards 2 as the solution is made very dilute
- for K_2SO_4, as expected, i approaches 3

TABLE 1.D *Values of van't Hoff factor, i, at Various Concentrations for NaCl, KCl, $MgSO_4$ and K_2SO_4.*

Salt	*Values of i			i if fully dissociated
	0.1 m	0.01 m	0.001 m	
NaCl	1.87	1.94	1.97	2.00
KCl	1.85	1.94	1.98	2.00
$MgSO_4$	1.21	1.53	1.82	2.00
K_2SO_4	2.32	2.70	2.84	3.00

* i values where dissociation is incomplete

Worked example degree of association (part 2)

$$1 - \frac{x}{2} = \frac{122\ \text{g mol}^{-1}}{241.98\ \text{g mol}^{-1}} = 0.504 \Rightarrow \frac{x}{2} = 1 - 0.504 = 0.496 \Rightarrow x = 2 \times 0.496 = 0.992$$

- benzoic acid is **99.2%** associated in benzene

Worked example van't Hoff factor and K_a of acetic acid

0.6 mL of acetic acid (CH_3COOH), density 1.06 g mL^{-1}, is dissolved in 1 litre of water, and the freezing point falls by 0.0205 °C. Find the van't Hoff factor and the acid's dissociation constant.

- moles of acetic acid = $\frac{0.6\ \text{mL} \times 1.06\ \text{g mL}^{-1}}{60\ \text{g mol}^{-1}} = 0.0106\ \text{mol} = n$
- molality = $\frac{0.0106\ \text{mol}}{1000\ \text{mL} \times 1\ \text{g mL}^{-1}} = 0.0106\ \text{mol kg}^{-1}$
- from $\Delta T_f = K_f\, m$: calculated $\Delta T_f = 1.86\ \text{K kg mol}^{-1} \times 0.0106\ \text{mol kg}^{-1} = 0.0197\ \text{K}$
- i = observed / calculated freezing point depression = 0.0205 K / 0.0197 K = **1.041**
- acetic acid is a weak electrolyte; each molecule that dissociates gives one acetate ion and one hydrogen ion
- with degree of dissociation x:

	CH_3COOH	$\rightleftharpoons H^+$	$+ CH_3COO^-$
initial	n mol	0	0
at equilibrium	$n(1-x)$ mol	nx mol	nx mol

- total particles: $n(1 - x + x + x) = n(1 + x)$
- $i = n(1 + x)/n = 1 + x = 1.041$, so $x = 1.041 - 1.000 =$ **0.041**
- $[CH_3COOH] = 0.0106\ (1 - 0.041)$; $[CH_3COO^-] = [H^+] = 0.0106 \times 0.041$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{0.0106 \times 0.041 \times 0.0106 \times 0.041}{0.0106\ (1.00 - 0.041)} = \mathbf{1.86 \times 10^{-5}}$$

Summary

Solutions and concentration

- a solution is a homogeneous mixture of two or more substances; solutions may be solid, liquid or gaseous
- concentration is stated as mole fraction, molarity, molality or percentages

Henry's and Raoult's laws

- Henry: at a given temperature, a gas's solubility in a liquid is directly proportional to its partial pressure
- Raoult: a non-volatile solute lowers the solvent's vapour pressure; the relative lowering equals the solute's mole fraction
- when both liquids of a binary solution are volatile: $p_{\text{total}} = p_1^{0}x_1 + p_2^{0}x_2$

Ideal, non-ideal, azeotropes

- ideal solutions obey Raoult's law at all concentrations
- deviations from it are positive or negative
- very large deviations give rise to azeotropes

Colligative properties

- depend on the number of solute particles, not their chemical identity: vapour-pressure lowering, boiling point elevation, freezing point depression, osmotic pressure
- osmosis reverses under a pressure above the osmotic pressure
- they are used to find solute molar masses

Abnormal molar masses and *i*

- dissociating solutes show molar masses below the true value; associating solutes show them above it
- van't Hoff factor *i* measures the extent: normal molar mass ÷ experimental molar mass, or observed ÷ calculated colligative property

Exercises

E1 What is a solution? How many types of solution are there? Describe each type briefly, with an example.

E2 Name a solid solution whose solute is a gas.

E3 Define: (i) mole fraction, (ii) molality, (iii) molarity, (iv) mass percentage.

E4 Laboratory concentrated nitric acid is an aqueous solution containing 68% nitric acid by mass, with a density of 1.504 g mL^{-1}. What is its molarity?

Exercises continued

E5 A glucose solution in water carries the label 10% w/w. Find the molality and the mole fraction of each component. Given a density of 1.2 g mL^{-1}, what is its molarity?

E6 A 1 g mixture of Na_2CO_3 and $NaHCO_3$ holds equal numbers of moles of each. What volume (mL) of 0.1 M HCl reacts with it completely?

E7 300 g of a 25% solution is mixed with 400 g of a 40% solution (both by mass). What is the mass percentage of the mixture?

E8 An antifreeze is made from 222.6 g of ethylene glycol ($C_2H_6O_2$) and 200 g of water. Find its molality, and its molarity if the density is 1.072 g mL^{-1}.

E9 Drinking water is badly contaminated with chloroform ($CHCl_3$), thought to be a carcinogen, at 15 ppm by mass. (i) Give this as a mass percentage. (ii) Find the molality of chloroform in the sample.

E10 How do molecular interactions matter in a solution of alcohol and water?

E11 Why does a gas always tend to become less soluble in a liquid as the temperature goes up?

E12 State Henry's law and give some of its important applications.

E13 Over a solution holding 6.56×10^{-3} g of ethane, ethane's partial pressure is 1 bar. What will the partial pressure be when the solution holds 5.00×10^{-2} g of ethane?

E14 What do we mean when a solution deviates positively, or negatively, from Raoult's law? How is the sign of $\Delta_{mix}H$ linked to each kind?

E15 A 2% aqueous solution of a non-volatile solute exerts 1.004 bar at the normal boiling point of the solvent. Find the solute's molar mass.

E16 Heptane and octane form an ideal solution; at 373 K their vapour pressures are 105.2 kPa and 46.8 kPa. What is the vapour pressure of a mixture of 26.0 g heptane and 35 g octane?

E17 Water has a vapour pressure of 12.3 kPa at 300 K. What is the vapour pressure of a 1 molal solution of a non-volatile solute in water?

E18 What mass of a non-volatile solute (molar mass 40 g mol^{-1}) must be dissolved in 114 g of octane to bring its vapour pressure down to 80%?

E19 30 g of a non-volatile solute in exactly 90 g of water gives a vapour pressure of 2.8 kPa at 298 K. After another 18 g of water is added, the vapour pressure is 2.9 kPa at 298 K. Find (i) the solute's molar mass and (ii) the vapour pressure of water at 298 K.

E20 5% (by mass) cane sugar in water freezes at 271 K. Pure water freezes at 273.15 K. At what temperature does 5% glucose in water freeze?

E21 Elements A and B form AB_2 and AB_4. In 20 g of benzene (C_6H_6), 1 g of AB_2 lowers the freezing point by 2.3 K and 1.0 g of AB_4 by 1.3 K. With benzene's molar depression constant = 5.1 K kg mol^{-1}, find the atomic masses of A and B.

Exercises continued

E22 At 300 K, a solution with 36 g of glucose per litre has an osmotic pressure of 4.98 bar. At the same temperature, what concentration gives an osmotic pressure of 1.52 bars?

E23 Name the chief kind of intermolecular attraction in each pair: (i) n-hexane and n-octane; (ii) I_2 and CCl_4; (iii) $NaClO_4$ and water; (iv) methanol and acetone; (v) acetonitrile (CH_3CN) and acetone (C_3H_6O).

E24 Using solute–solvent interactions, put cyclohexane, KCl, CH_3OH and CH_3CN in order of increasing solubility in n-octane, and give reasons.

E25 Sort into insoluble, partly soluble and highly soluble in water: (i) phenol, (ii) toluene, (iii) formic acid, (iv) ethylene glycol, (v) chloroform, (vi) pentanol.

E26 Lake water of density 1.25 g mL^{-1} holds 92 g of Na^+ ions per kg of water. What is the molarity of Na^+ in the lake?

E27 CuS has a solubility product of 6×10^{-16}. What is the greatest molarity CuS can reach in water?

E28 6.5 g of aspirin ($C_9H_8O_4$) is dissolved in 450 g of acetonitrile (CH_3CN). What is the mass percentage of aspirin?

E29 Nalorphene ($C_{19}H_{21}NO_3$), which resembles morphine, helps narcotic users through withdrawal; the usual dose is 1.5 mg. What mass of a 1.5×10^{-3} *m* aqueous solution supplies this dose?

E30 How much benzoic acid (C_6H_5COOH) is needed to make 250 mL of a 0.15 M solution in methanol?

E31 Equal amounts of acetic acid, trichloroacetic acid and trifluoroacetic acid depress the freezing point of water increasingly, in that order. Give a brief explanation.

E32 10 g of $CH_3CH_2CHClCOOH$ is added to 250 g of water. With $K_a = 1.4 \times 10^{-3}$ and $K_f = 1.86$ K kg mol^{-1}, find the depression of freezing point.

E33 Dissolving 19.5 g of CH_2FCOOH in 500 g of water depresses the freezing point by 1.00 °C. Find the van't Hoff factor and the dissociation constant of fluoroacetic acid.

E34 Water's vapour pressure at 293 K is 17.535 mm Hg. What is it at 293 K after 25 g of glucose is dissolved in 450 g of water?

E35 At 298 K the Henry's law constant for the molality of methane in benzene is 4.27×10^5 mm Hg. What is methane's solubility in benzene at 298 K under 760 mm Hg?

E36 100 g of liquid A (molar mass 140 g mol^{-1}) is dissolved in 1000 g of liquid B (molar mass 180 g mol^{-1}). Pure B has a vapour pressure of 500 torr and the solution's total vapour pressure is 475 Torr. Find the vapour pressure of pure A, and of A in the solution.

Exercises continued

E37 At 328 K pure acetone and pure chloroform have vapour pressures of 741.8 mm Hg and 632.8 mm Hg. Treating their mixtures as ideal over all compositions, plot p_{total}, $p_{chloroform}$ and $p_{acetone}$ against $x_{acetone}$. On the same graph paper, plot the measured values below, and say whether the mixture deviates positively or negatively from ideal behaviour.

$100 \times x_{acetone}$	0	11.8	23.4	36.0	50.8	58.2	64.5	72.1
$p_{acetone}$ / mm Hg	0	54.9	110.1	202.4	322.7	405.9	454.1	521.1
$p_{chloroform}$ / mm Hg	632.8	548.1	469.4	359.7	257.7	193.6	161.2	120.7

E38 Benzene and toluene form an ideal solution at all compositions; at 300 K their pure vapour pressures are 50.71 mm Hg and 32.06 mm Hg. For 80 g of benzene mixed with 100 g of toluene, find benzene's mole fraction in the vapour.

E39 Air is a mixture of gases, chiefly oxygen and nitrogen in roughly 20 : 79 by volume at 298 K. Water is in equilibrium with air at 10 atm. At 298 K the Henry's law constants are 3.30×10^7 mm for oxygen and 6.51×10^7 mm for nitrogen. Find the make-up of these gases dissolved in the water.

E40 What amount of $CaCl_2$ (i = 2.47) must be dissolved in 2.5 litre of water to give an osmotic pressure of 0.75 atm at 27 °C?

E41 25 mg of K_2SO_4 is dissolved in 2 litre of water at 25 °C. Taking it as fully dissociated, find the osmotic pressure.

Answers to some intext questions

Question	Answer	Question	Answer
Q1	C_6H_6 = 15.28%, CCl_4 = 84.72%	Q5	1.5 mol kg^{-1}, 1.45 mol L^{-1}, 0.0263
Q2	0.459, 0.541	Q9	23.4 mm Hg
Q3	0.024 M, 0.03 M	Q10	121.67 g
Q4	36.946 g	Q11	5.077 g
		Q12	30.96 Pa